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Can Approval-Day Information Predict Interruptions of Conditional Lending Programmes?

A Comparison of Random Forests, Logistic Regression and Neural Networks

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Course: 5,155,1.00 Machine Learning in Finance

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1 INTRODUCTION AND RELATED LITERATURE

1.1 Motivation and financial relevance

Many official lenders provide conditional financing: they disburse a loan in tranches and release each tranche after the first only when a review confirms that the borrower is meeting the policy conditions agreed at approval. The World Bank's development policy financing and the European Union's macro-financial assistance, for example, also attach policy conditions to their loans. We study interruption risk using programmes of the International Monetary Fund (IMF), which form a natural sample: the IMF's Monitoring of Fund Arrangements database (MONA) records the review schedule of IMF-supported programmes approved since 2002 and whether each review was completed (IMF, 2026b), which gives a consistent record at the programme level.

The IMF supports countries facing balance-of-payments difficulties through lending programmes that combine financing with policy commitments and periodic reviews. We ask whether information available at approval by the IMF Executive Board (the Board) can predict whether a programme is later interrupted (see Section 2.1 for the outcome definition).

Interruption matters financially because implementation can affect expected financing and confidence in the government's policy commitments. Mody and Saravia (2006, pp. 843–845, 857) find that IMF programmes do not uniformly improve access to international bond markets; favourable effects depend on credible reform commitments and economic circumstances. Reinsberg et al. (2022, pp. 1023, 1034–1035) associate permanent interruptions with lower institutional investor ratings and a short-lived increase in sovereign bond spreads.

1.2 Political incentives and implementation capacity

The implementation literature emphasises the borrowing government's willingness and ability to fulfil its commitments. Ivanova et al. (2003, pp. 4, 34) identify political cohesion, stability and bureaucratic quality as important influences on implementation. Joyce (2006, pp. 342–345, 351) similarly links implementation to government ideological cohesion, regime duration and political and economic openness. Dreher (2003, pp. 101, 111–114) shows that electoral incentives interact with political institutions and also identifies associations with initial economic vulnerability.

Our 13 inputs do not directly measure elections, political cohesion or bureaucratic quality, so these mechanisms are only partly observed.

1.3 Conditionality and programme design

Evidence on conditionality is mixed. Ivanova et al. (2003, pp. 4, 34) find limited explanatory power for conditionality once domestic political factors are considered. Dreher's (2009, pp. 248–250) survey discusses competing arguments about conditions as commitment devices and constraints on implementation. More recently, Reinsberg et al. (2022, pp. 1023, 1026, 1035) find that greater structural conditionality at programme onset is associated with interruption, especially conditions involving public-sector reform, privatisation and price liberalisation.

The outcome definition also affects interpretation, because duration and review counts may predict interruption partly through exposure (Section 3.2). Counting conditions also misses their political difficulty and content.

1.4 Macroeconomic vulnerability and prior programme use

Weak growth, inflation and external financing pressures can make agreed adjustment harder to implement. Iseringhausen et al. (2019, pp. 8, 16, 19, 23–25) find that macroeconomic conditions, institutions and previous IMF experience help explain repeated programme use. Their outcome, however, is a successor arrangement within three years. A country can return to the IMF after successfully completing a programme because it experiences a new shock.

1.5 Machine learning in related work and our contribution

Machine learning already contributes to IMF forecasting. Agbloyor et al. (2023, pp. 1–5) predict IMF bailouts using 6,550 observations and 138 features, reporting predictive information from economic, agricultural, energy, health and social indicators. Batsuuri et al. (2024, pp. 12, 14–18, 22–25) find that machine-learning methods outperform traditional approaches when forecasting new IMF arrangements, with results sensitive to preprocessing and prediction horizon. Both studies predict participation; we predict what happens after a programme is approved.

Gu et al. (2020, pp. 2223, 2225–2227) demonstrate the value of non-linearities and interactions in equity-return prediction. Their large stock-return panel differs from our 173-programme sample, so gains from flexibility may not carry over; logistic regression provides a benchmark. Breiman (2001a, pp. 5–8, 10–11) provides the basis for random forests, which combine predictions from randomised trees. We compare the random forest with logistic regression, an interpretable benchmark for a binary outcome, and a neural network.

Our contribution is a focused predictive test, in the sense of Shmueli (2010), of the research question stated in Section 2.1. The literature covers implementation mechanisms, financial consequences, predictors of participation or repeated use, and the determinants of interruption, but to our knowledge no study evaluates out-of-sample prediction of interruption with approval-day information.

2 RESEARCH DESIGN

2.1 Research question and prediction task

The general question of this study is whether the information available on the day a conditional lending programme is approved can predict whether the programme will later be interrupted, and whether a flexible model such as a random forest predicts this better than logistic regression or a neural network. We use IMF programmes as the sample for this analysis, because MONA records the scheduled and completed reviews of IMF-supported programmes approved since 2002 (Section 1.1). The unit of observation is one lending programme. The prediction is made at Board approval, using only data known on that date.

This is a supervised binary classification task. The output equals 1 when fewer reviews were completed than scheduled at approval. It equals 0 when every scheduled review was completed. The inputs are 13 variables in four groups: deal structure, conditionality, track record and macroeconomic conditions. Section 3.2 lists them.

Following James et al. (2013, pp. 16–17), we estimate f in $\Pr(Y = 1 \mid X) = f(X)$. The estimate is judged on programmes it has not seen during training. Following Shmueli (2010, p. 291), we treat this as a prediction problem. Coefficients and variable importance are not read as causal effects.

Performance is measured on a later test set, with interrupted programmes as the positive class. We report accuracy; sensitivity, the share of interrupted programmes the model flags; specificity, the share of completed

programmes it correctly clears; and balanced accuracy, the mean of sensitivity and specificity (Brodersen et al., 2010, p. 3121). We also report the area under the receiver operating characteristic curve (AUC), the probability that a model ranks a randomly chosen interrupted programme above a randomly chosen completed one, so 0.500 corresponds to random guessing (Fawcett, 2006, p. 868). Interruptions are the majority in the test set (see Section 3.1). A rule that always predicts interruption therefore gets high accuracy. Its balanced accuracy is only 0.500. Plain accuracy misleads when classes are unequal (He & Garcia, 2009, p. 1276). Balanced accuracy removes this advantage. The random forest and the neural network are tuned on balanced accuracy for the same reason; the logistic regression is not tuned.

An approval-day flag could, if accurate, help the IMF and creditors assess the risk of reviews being left uncompleted and subsequent disbursements being withheld (IMF, n.d.; Section 6).

2.2 Study design, data sources and train/test split

The study is a cross-section of IMF programmes. Each row is one programme, described as it stood at approval. We keep facilities with scheduled reviews, including their predecessors. These are the Stand-By Arrangement (SBA), the Extended Fund Facility (EFF), the Extended Credit Facility (ECF), and the Standby Credit Facility (SCF). The ECF and the SCF are concessional facilities of the IMF's Poverty Reduction and Growth Trust (PRGT). Facilities without reviews and precautionary arrangements are dropped (see Section 3.1). Approvals run from 2002 to 2016. Review records in MONA start in 2002. Ending the window in 2016 limits overlap with the COVID-19 period.

No single public source records programme design and review outcomes together. We therefore built the dataset from IMF files. The IMF Financial Data Query Tool gives the history of lending arrangements (IMF, 2026a); we call this source the commitments file. It provides dates, facility type, and access in percent of quota at approval. MONA covers reviews, conditions, and staff macro estimates (IMF, 2026b). We use four MONA files. Reviews lists scheduled and completed reviews, from which we build the output. Combined lists structural conditions and prior actions. Mecon holds staff estimates of growth, inflation, and the current account. Description holds a cancellation flag, used only to check the output. The final macro inputs come from MONA; Section 3.2 explains why.

The commitments file and MONA share no common identifier. Country names are spelled differently, and approval dates can differ by a few days. We normalise country names and match on the closest approval date within 60 days. Blends, in which two facilities (for example an ECF and an EFF) are approved for the same country on the same day, become one programme, because they share one review schedule. Overlapping or successive programmes of the same country are kept as separate observations (Section 6). Section 3.1 reports the resulting sample.

We split the sample by approval year. Training covers 2002 to 2013, and the test set covers 2014 to 2016. The test programmes are never used to fit or tune the models. All inputs are taken as they stood on approval day (see Section 3.2). Hyperparameters are tuned on the training set with 10-fold cross-validation (CV) and leave-one-out cross-validation (LOOCV). A third scheme, country-grouped 10-fold CV (country CV), holds out whole countries in each fold (Roberts et al., 2017, p. 914). Many countries have several programmes, so random folds put one country on both sides. We use the country-grouped scores to choose the tuned settings for the random forest and the neural network. A prediction for a programme from a model fitted without that programme is called an out-of-fold prediction.

3 DATA

3.1 Sample construction and summary statistics

Our final dataset covers 173 IMF lending programmes in 86 countries, approved between 2002 and 2016; 82 of them (47.4%) were interrupted.

Starting from the 1,576 lending arrangements since 1952 in the commitments file, we keep the 221 non-precautionary arrangements approved in 2002–2016. We drop rapid one-off loans under the Rapid Credit Facility (RCF) and the Rapid Financing Instrument (RFI), which have no review schedule and so cannot be interrupted, and the Precautionary and Liquidity Line (PLL), a precautionary instrument. Merging blended programmes into one observation leaves 182 programmes, of which 173 match MONA. We split by approval year (training 2002–2013, $n = 145$; test 2014–2016, $n = 28$) and fill the few missing values with training-set medians, so nothing from the test period leaks into training. Table 1 summarises the 13 inputs.

The classes are close to balanced overall, but interruptions rise from 39% of approvals in 2002–2004 to 61% in 2014–2016. They are the minority in training (65 of 145, 44.8%) and the majority in the test set (17 of 28, 60.7%). Always predicting "interrupted" would therefore already be right for 60.7% of test programmes, which is the accuracy benchmark; on balanced accuracy, the metric we use, this rule scores 0.500.

ECFs, concessional loans to low-income countries, make up 97 of the 173 programmes, followed by 47 SBAs, 16 EFFs, 8 blends and 5 SCFs. Interruption rates range from 64% for SBAs to 38% for ECFs and blends.

A typical programme runs for three years with six scheduled reviews, ten structural conditions and two prior actions. Access in percent of quota is strongly right-skewed, so we use its log, and the macro variables have long tails (inflation runs from −18% to 48%). A few extreme macro values (e.g. Yemen's 2014 inflation of −17.7%) may reflect recording issues in MONA. Missing values are rare: the time since the previous programme is undefined for the 11 first-time borrowers, and no other input misses more than 10 values. Each missing value is filled with the median of the 145 training programmes, and the test set uses the same training medians. For the 11 first-time borrowers, the time since the previous programme is set to the training median of 1.6 years; the no-track-record indicator equals 1 for exactly these programmes, so the models can distinguish them from countries that returned after a short gap. The medians are computed once on the full training set, so the cross-validation folds share them; with at most 11 missing values per input, this effect on the cross-validation scores is small.

Across all 173 programmes, interrupted programmes differ from completed ones mainly in deal structure, although planned length differs little. They schedule more reviews (7.3 against 5.9), receive more access relative to quota (mean log access 4.96 against 4.31) and are less often concessional (52% against 74%) or extended (59% against 79%), all with Welch t-test p-values below 0.01. Gaps in conditionality, track record and macroeconomic conditions are smaller, with Welch t-test p-values of 0.09 or more.

3.2 Output and input variables

The output is whether a programme was interrupted. All 13 inputs are known on the day the Board approves the programme, which is when the prediction is made.

Selection principle. A predictive input must be available at the time of prediction and carry information about the outcome; it needs no causal interpretation (Shmueli, 2010). We therefore use only approval-day information: the "upon approval" columns of the commitments file and MONA's approval snapshot (R0).

Table 1. Summary statistics of the 13 inputs (before imputation, n = 173)

Variable	Unit	Missing	Mean	SD	Min	Median	Max
log_access_pct_quota	log of % of quota	0	4.62	1.29	1.61	4.50	8.08
planned_months	months	0	32.58	7.56	7.20	35.98	47.97
n_reviews_sched	count	0	6.60	2.27	2.00	6.00	16.00
n_struct_cond	count	1	10.67	4.98	2.00	10.00	26.00
n_prior_actions	count	1	2.79	2.98	0.00	2.00	16.00
prev_n_20y	count	0	3.35	2.06	0.00	3.00	8.00
years_since_prev	years	11	4.09	6.82	0.00	1.62	47.74
growth	%	5	4.04	4.23	−8.69	4.15	24.46
inflation	%	8	7.73	7.93	−17.69	5.50	48.04
current_account_gdp	% of GDP	10	−8.62	11.77	−92.49	−6.20	15.36
concessional	0/1	0	0.64	0.48	0.00	1.00	1.00
extended	0/1	0	0.69	0.46	0.00	1.00	1.00
no_track_record	0/1	0	0.06	0.25	0.00	0.00	1.00

Note. For binary variables the mean is the share of programmes with value 1. Variable names are explained in Table 2. Source: authors' calculations from IMF MONA and lending-arrangements data.

Table 2. Input and output variables

Variable	Group	Description	Source
interrupted	Output	1 = fewer reviews completed than scheduled at approval	MONA Reviews
log_access_pct_quota	Deal structure	Log of agreed access, % of IMF quota	Commitments file, upon approval
planned_months	Deal structure	Planned length, approval to scheduled expiry	Commitments file, upon approval
concessional	Deal structure	1 = includes a concessional (PRGT) facility	Commitments file
extended	Deal structure	1 = EFF or ECF	Commitments file
n_reviews_sched	Deal structure	Reviews scheduled at approval	MONA Reviews, R0
n_struct_cond	Conditionality	Structural conditions at approval	MONA Combined, R0
n_prior_actions	Conditionality	Prior actions before Board approval	MONA Combined, R0
prev_n_20y	Track record	Programmes in the 20 years before approval	Commitments file, history
years_since_prev	Track record	Years since the previous programme's planned end (0 if overlapping)	Commitments file, history
no_track_record	Track record	1 = no earlier programme	Commitments file, history
growth	Macroeconomic	Real GDP growth, %, staff estimate for the approval year	MONA Mecon, R0
inflation	Macroeconomic	CPI inflation, %, staff estimate for the approval year	MONA Mecon, R0
current_account_gdp	Macroeconomic	Current-account balance, % of GDP	MONA Mecon, R0

Note. R0 = MONA's approval snapshot. PRGT = Poverty Reduction and Growth Trust. EFF = Extended Fund Facility; ECF = Extended Credit Facility.

Output. Tying the label to the approval-day review schedule means later rescheduling cannot change it. MONA's cancellation flag would be too narrow: only 29 of the 82 interrupted programmes were formally cancelled, while the other 53 ended with reviews outstanding.

Inputs. Table 2 groups the inputs into deal structure, conditionality, track record and macroeconomic conditions.

Transformations and overlap. Access enters in logs. Facility type is left out because the concessional and extended indicators identify it exactly, and the number of tranches because it is almost always the number of reviews plus one. Some inputs remain correlated (the extended indicator with planned length, $r = 0.77$; the extended indicator with the concessional indicator, $r = 0.81$), which matters little for the forest's predictions but makes the importance of each input and the logistic regression coefficients harder to read.

Excluded variables. Post-approval outcomes and identifiers are excluded, as are fiscal variables: the April 2026 World Economic Outlook contains revised rather than approval-day values (IMF, 2026c), and MONA lacks them for 24–45% of programmes. The full exclusions are listed in the supplementary Excel file submitted with this report.

Caveats. The number of scheduled reviews enters both an input and the outcome definition, so part of its predictive power is mechanical. The macro inputs are staff projections at approval, which may be optimistic but are what was known at the time.

4 RANDOM FOREST

4.1 Method, tuning and test predictions

Method. The first model is a random forest. Because our study is a classification problem, a random forest appeared a promising starting point (Batsuuri et al., 2024, p. 3). This model combines many classification trees, each splitting the data through a series of simple yes/no questions.

A single tree captures non-linear patterns and interactions, but it is unstable: small changes in the data can produce a very different tree. For this reason, we use bagging (bootstrap aggregation). The idea is to grow many trees, each on a bootstrap sample (programmes drawn with replacement), and average their votes. Averaging helps to cancel out noisy predictions. However, bagging alone has a weakness. If one input is very strong, almost every tree will split on it first, making the trees similar, which limits the gain from averaging (James et al., 2013, p. 320). Indeed, Hastie et al. (2009, p. 588) formalised it with the variance of an average of B trees, each with variance σ^2 and pairwise correlation ρ:

$$\text{Var} = \rho\sigma^2 + \frac{(1-\rho)\sigma^2}{B} \tag{1}$$

More trees (B) shrink the second term, but the first term only shrinks with less correlated trees. We use Breiman's (2001a, p. 6) solution of limiting each split to a random subset of *mtry* inputs. Each tree then votes: the share of trees voting "interrupted" gives the predicted probability, and a threshold assigns the class. The programmes left out of each bootstrap sample (out-of-bag, OOB) provide a built-in error estimate.

Default forest. The default forest was trained with the default hyperparameters of the *randomForest* R package (Liaw & Wiener, 2002): 500 trees, *mtry* 3. Its OOB balanced accuracy is 0.609.

Hyperparameter tuning. To tune our hyperparameters (number of trees and *mtry*), we scored 24 different pairs on balanced accuracy (Section 2.1), ranging from 100 to 1000 trees and *mtry* from 2 to 12, with the *caret* package (Kuhn, 2008).

We used three different cross-validation schemes: 10-fold CV, LOOCV and country-grouped 10-fold CV (country CV, Section 2.2).

For 10-fold CV and country CV, we compute balanced accuracy within each fold and average over the ten folds; for LOOCV, where each fold holds a single programme, we compute it once over all out-of-fold predictions. Table 3 compares the default forest with the best pair under each scheme.

Validation scheme	**Default forest (500 trees, *mtry* 3)**	**Best pair under this scheme**	**Balanced accuracy of best pair**
Out-of-bag (default forest only)	0.609	n/a	n/a
10-fold CV	0.623	250 trees, *mtry* 2	0.657
Country-grouped 10-fold CV	0.597	250 trees, *mtry* 3	0.620
LOOCV	0.606	100 trees, *mtry* 8	0.635

Table 3. Default against tuned forest on the training set (balanced accuracy; best of 24 pairs per scheme)

The default pair (500, 3) is part of the tuning grid, so its CV scores come from the same runs. OOB accuracy of the default forest is 0.628.

Country CV gives the lowest best score (0.620, against 0.657 for 10-fold CV and 0.635 for LOOCV).

Final model. We keep the pair chosen under country CV (250 trees, *mtry* 3), refit it on all 145 training programmes with the threshold that maximises out-of-fold balanced accuracy (0.51), and predict the 28 test programmes once (Table 4). The selected pair keeps the default *mtry* and only lowers the number of trees from 500 to 250, which cannot be expected to improve accuracy, so its gain over the default under country CV (0.620 against 0.597) most likely reflects noise in a search over 24 pairs. Only the test-set rows of Table 4 measure out-of-sample performance; the other figures in Tables 3 and 4 are training-set estimates used to choose settings.

Out-of-sample predictions. On the 28 test programmes, the final forest reaches a balanced accuracy of 0.508, barely above the 0.500 of always predicting "interrupted", and its accuracy (0.500) is even below that rule's (0.607); the rule's high accuracy despite zero skill shows why plain accuracy misleads when classes are unequal. Out of fold on the training set, it reaches 0.630; this is more precise than the test score but slightly optimistic, since both the pair and the threshold were chosen on these predictions. The default forest's higher test balanced accuracy (0.567 against 0.508) comes entirely from two interrupted programmes with a vote share of 0.504: they are classed as interrupted at threshold 0.50 but as completed at 0.51, and at 0.50 the final forest also scores 0.567. The default forest also ranks programmes better (AUC 0.620 against 0.586), so tuning did not improve the test results on either measure. On the test set, one extra correct prediction moves balanced accuracy by about 0.03 to 0.05. The forest thus finds a modest signal in the training set, but no reliable advantage over chance on later programmes.

Model	Accuracy	Sensitivity	Specificity	Balanced accuracy	AUC
Baseline (always interrupted)	0.607	1.000	0.000	0.500	0.500
Default forest (threshold 0.50)	0.571	0.588	0.545	0.567	0.620
Final forest (threshold 0.51)	0.500	0.471	0.545	0.508	0.586
Final forest (threshold 0.50)	0.571	0.588	0.545	0.567	0.586
Final forest, training out-of-fold (n = 145, threshold 0.51)	0.641	0.523	0.738	0.630	0.631

Table 4. Random forest on the test set (28 programmes, 2014–2016), with the final forest's training out-of-fold scores for reference

Note. Final forest: 250 trees, *mtry* 3. The 0.50 row uses the same threshold as the logistic regression, for a like-for-like comparison; it differs from 0.51 only because two interrupted programmes sit at a vote share of 0.504. The last row is computed on the 145 training programmes, not on the test set.

4.2 Variable importance

Permutation importance shuffles the values of each input among the out-of-bag samples, breaking its link with the target, and measures the resulting drop in accuracy, averaged over all trees. Higher values mean higher predictive power, while small or negative values signal no contribution. The mean decrease in accuracy (MDA) is one of the two importance measures in *randomForest*, the other being the mean decrease in Gini impurity (Abedin et al., 2021, p. 23). Importance indicates neither direction nor causality, and correlated inputs (the extended and concessional indicators) share credit.

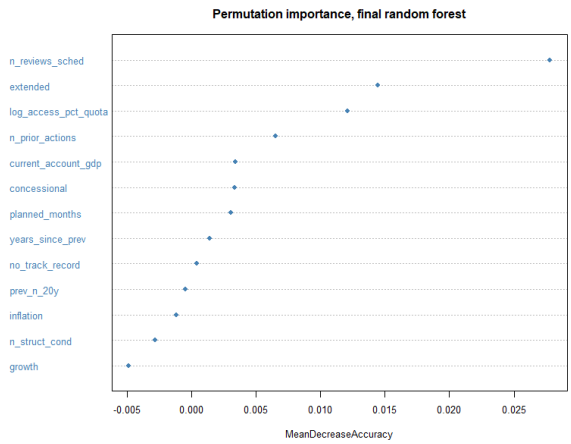


Figure 1. Permutation importance of the final forest.

Note. Mean decrease in out-of-bag accuracy when one input is shuffled. Higher values mean more important; values near or below zero mean no contribution.

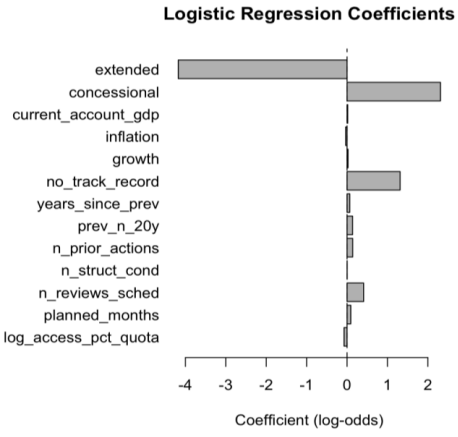


Figure 2. Logistic regression coefficients for programme interruption.

Note. Coefficients are in log-odds and estimated on the training set. Positive values indicate higher odds of interruption. Inputs are on different scales, so bar lengths are not directly comparable measures of importance. Extended and concessional are strongly correlated (r = 0.81).

The number of scheduled reviews is the strongest input (Figure 1): 85% of training programmes with eight or more reviews were interrupted, against 36% of the others. Part of this is mechanical, as each extra review is

one more chance to miss one (Section 3.2). Because this link is partly built into the outcome, the forest's training-set signal should be read as an upper bound on what approval-day information reveals about implementation. Programme length receives little importance on its own because it overlaps with the extended indicator (r = 0.77); within each facility type, longer training programmes were interrupted more often (all 10 non-extended programmes planned for more than 30 months, against 49% of those planned for 30 months or less), in line with Reinsberg et al. (2022, p. 1028), who find that longer agreements fail more often. Extended facilities (EFF or ECF) come second, with 38% of extended training programmes interrupted against 60% of the others, in line with the IMF's review: "EFF and EFF-ECF arrangements, generally longer in duration than SBAs, had higher completion rates" (IMF, 2019, p. 45). In our sample the gap mainly reflects ECFs (36% of training ECFs interrupted), which are also concessional; EFFs alone (57%) differ little from SBAs (60%). Section 6 discusses whether higher completion in extended programmes reflects better implementation. Access relative to quota ranks third and prior actions fourth (48% interrupted with at least one prior action, against 37% without), in line with the IMF's finding of "a negative association between PAs [prior actions] at arrangement approval and completion rates" (IMF, 2019, p. 45).

Growth and inflation add no predictive value here; staff projections may be optimistic, and the literature gives mixed findings on initial economic conditions (Ivanova et al., 2003, p. 1; Reinsberg et al., 2022, p. 1028; Dreher, 2003). A count of structural conditions also adds no predictive value, but it cannot test Reinsberg et al.'s (2022, p. 1028) finding on specific condition types.

5 ALTERNATIVE MODELS

5.1 Logistic regression

To evaluate an alternative to the random forest, we used a logistic regression model to predict whether an IMF lending programme would be interrupted. We chose logistic regression because the output is binary (Hosmer et al., 2013, p. 1; James et al., 2013, pp. 130–131). It provides a relatively simple and interpretable benchmark against which the performance of a more complex model can be analysed.

On the test set of 28 programmes, logistic regression at threshold 0.50 achieved an accuracy of 0.536, a balanced accuracy of 0.570 and an AUC of 0.684, against 0.500, 0.508 and 0.586 for the random forest at its selected threshold of 0.51. At the same 0.50 threshold, however, the forest reaches 0.571 and 0.567, so the two models differ mainly in AUC, which does not depend on the threshold (Fawcett, 2006, p. 868): logistic regression ranks interrupted programmes above completed ones more consistently. With 28 test programmes, even this gap should be read cautiously, and both results show that predicting interruptions remains difficult. Table 5 in Section 5.4 lists all models side by side.

A key advantage of logistic regression is its interpretability. Its coefficients indicate how each input is associated with the log-odds of interruption, holding the other inputs constant (Hosmer et al., 2013, pp. 49–52; see Figure 2). The largest coefficients are on the two binary facility indicators, extended (−4.18, p = 0.004) and concessional (2.31, p = 0.087), of which only the former is statistically significant at the 5% level. These two inputs are strongly correlated in the training data (r = 0.81). As a result, the model splits their joint association into opposite signs and estimates both with large standard errors, so their individual coefficients should not be read in isolation (James et al., 2013, pp. 99–102). The absence of a track record also has a sizeable positive coefficient (1.31), but it is imprecisely estimated (p = 0.231). More generally, these associations are not causal,

and coefficient magnitudes are not directly comparable because the inputs are measured on different scales (Hosmer et al., 2013, pp. 62–64).

The forest combines randomised trees (Breiman, 2001a, pp. 5–6) and captures non-linear relationships and interactions (Couronné et al., 2018, p. 4), at the cost of interpretability (Breiman, 2001b, p. 208), while logistic regression's simpler form is more transparent (James et al., 2013, pp. 24–26). At the common 0.50 threshold, logistic regression matched the forest's balanced accuracy and ranked programmes better, but 28 test programmes warrant caution; complexity does not guarantee better predictions (Hastie et al., 2009, p. 221; Christodoulou et al., 2019, accepted manuscript, pp. 2–3; Couronné et al., 2018, pp. 7–9), and interpretability remains important (Rudin, 2019, pp. 206–207).

5.2 Neural network: architecture and preprocessing

As a second alternative, we fit a neural network. A feedforward neural network passes the inputs through layers of neurons, each of which applies a non-linear activation function, such as the sigmoid, to a weighted sum of its inputs. This non-linearity is what lets the network learn non-linear relationships (Wüthrich & Merz, 2023, pp. 269–273). We use the R package *nnet*, which fits networks with a single hidden layer and optimises the weights with the Broyden–Fletcher–Goldfarb–Shanno (BFGS) method (Ripley & Venables, 2026; Venables & Ripley, 2002). Since flexible networks can also fit the noise in the training data, we keep the network small and add weight decay, which can improve generalisation (Krogh & Hertz, 1992, p. 950). The 13 inputs are rescaled to [0, 1], as inputs on similar scales help the weights learn at a similar rate (LeCun et al., 1998, Sect. 4.3). The minimum and maximum come from the training set only, so that no information from the test set leaks into the model (Kapoor & Narayanan, 2023, pp. 1–3).

5.3 Neural network: tuning and threshold selection

We tune the number of hidden neurons (1, 2, 3 or 5) and the weight decay (0.01, 0.1, 0.5 or 1) with the *caret* package (Kuhn, 2008). As for the random forest, we use country-grouped 10-fold CV (Section 2.2). Models are compared by balanced accuracy (Section 2.1).

The best network has two hidden neurons and a decay of 0.01, with a cross-validated balanced accuracy of 0.650. With the same light decay, five neurons only reach 0.544, while a strong decay of 1 keeps every size between 0.510 and 0.530, close to the 0.500 of a model that always predicts the same class. With 145 training programmes, a small and lightly regularised network therefore works best. We then choose the classification threshold that maximises balanced accuracy on the out-of-fold predictions, so that the test set plays no role in model selection. This gives a threshold of 0.44 and an out-of-fold balanced accuracy of 0.684.

5.4 Test-set comparison of all models

Table 5 compares the final network (13 inputs, 2 hidden neurons, 31 weights) with the other models on the 28 test programmes approved between 2014 and 2016, and states the classification threshold behind each row. The metrics are defined in Section 2.1.

At their selected thresholds, the network beats the random forest on balanced accuracy (0.556 against 0.508) but not on AUC (0.567 against 0.586), and both reach an accuracy of only 0.500, below the 0.607 of predicting an interruption for every programme. At a common threshold of 0.50, the three models reach balanced accuracies of 0.567 (forest), 0.570 (logistic regression) and 0.572 (network), within 0.005 of each other. The two models

Table 5: Test-set performance of all models (28 programmes, 2014–2016)

Model	Threshold	Accuracy	Sensitivity	Specificity	Balanced accuracy	AUC
Neural network	0.44	0.500	0.294	0.818	0.556	0.567
Final forest	0.51	0.500	0.471	0.545	0.508	0.586
Final forest	0.50	0.571	0.588	0.545	0.567	0.586
Logistic regression	0.50	0.536	0.412	0.727	0.570	0.684
Baseline (always interrupted)	n/a	0.607	1.000	0.000	0.500	0.500

Note. Thresholds for the network (0.44) and the final forest (0.51) were chosen on training out-of-fold predictions; the logistic regression uses 0.50. At 0.50 the network's balanced accuracy is 0.572.

also make different errors: at their selected thresholds of 0.44 and 0.51, respectively, the network rarely predicts an interruption (sensitivity 0.294, specificity 0.818), while the forest is more balanced (0.471 and 0.545). The network's result is also fragile. Refitting it with 20 different random seeds gives test balanced accuracies between 0.436 and 0.559 (median 0.553), so the 0.556 in Table 5 sits near the top of a range that reaches down to 0.436. This is expected, since the starting weights are drawn randomly (Wüthrich & Merz, 2023, p. 284) and the fitted network is not unique.

We thus find no reliable winner between the two models. The network's balanced accuracy also falls from 0.684 out of fold, which is optimistic because the settings and the threshold were chosen on those predictions (cross-validated 0.650), to 0.556 on the later test set. Since it brings no clear accuracy gain over the other models, we would not use the neural network to flag programmes at risk of interruption.

6 DISCUSSION AND CONCLUSION

Overall, approval-day information predicts interruption only weakly, and no model is reliably better. On the 28 test programmes, balanced accuracy ranges from 0.51 to 0.57 at the model-specific thresholds in Table 5, against 0.500 for the baseline. At a common threshold of 0.50, the forest (0.567), logistic regression (0.570) and network (0.572) differ by 0.005, while one additional correct prediction moves balanced accuracy by about 0.03 to 0.05. Logistic regression has the highest AUC (0.684).

Several limitations qualify these findings, beginning with the treatment of parallel and successive programmes. Same-country, same-day blends share a review schedule and are merged; overlapping and successive programmes remain separate observations, with history captured by track-record inputs. There are 13 programmes overlapping earlier ones (11 in training), and 43 of the 84 training countries have several programmes. Cancelled programmes replaced by successors count as interrupted, including some superseded arrangements. Section 2.2 describes country CV; 26 of the 28 test programmes come from training countries, so the test mainly assesses known borrowers. In addition, the test set is small (28 programmes), its programmes are larger, longer and less often concessional, scheduled reviews partly define the outcome, and the macro inputs are staff projections.

From a policy perspective, these scores could complement staff judgement rather than flag programmes independently; the mechanical role of scheduled reviews does not justify scheduling fewer reviews. The findings for extended programmes also need careful interpretation: they were interrupted less often (38% against 60% in training), mainly because of ECFs (36%); EFFs (57%) resembled SBAs (60%). Long relationships can create moral hazard: expected waivers, modified targets or lenient reviews weaken borrower incentives, while lenders may hesitate to interrupt. The IMF's Independent Evaluation Office examined prolonged use, and Reinsberg et al. (2022) use "dependency trap" in their title. Completed reviews cannot distinguish compliance from leniency, so high completion alone is weak evidence of success.

For further research, one avenue would be to use pre-2002 review histories from other IMF sources or completed post-2016 programmes, assessed with rolling time-ordered splits. Re-estimating the models without the number of scheduled reviews would measure the mechanical part of the signal.

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DECLARATION OF AI USE

List of Aids

Resource	Use	Affected Section
EndNote	Reference and citation management	Entire report
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Claude (Anthropic)	Help with correcting and designing code, including R code debugging; literature search; review and editing of drafts, including grammar, spelling and rephrasing.	Entire report + code

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